NIHAL HAMOU-SBIRA KTITU

Data Scientist | Machine Learning Engineer
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SUMMARY

Data scientist and machine learning engineer with a software engineering background and a Master's in Data Science & AI (9.2/10 average, 10/10 thesis). I build end-to-end machine learning systems, from raw data to deployed apps: statistical modeling, predictive models, model monitoring, MLOps and LLM applications, focused on financial services.

EDUCATION

Master's in Data Science & Artificial Intelligence – Evolve, Madrid Jan 2026 – Sep 2026

- Average grade: 9.2/10 · Master's thesis: 10/10

Higher Technical Degree in Multiplatform App Development (DAM) – EEP iGroup, Madrid Sep 2022 – Jul 2024

- Average grade: 8.66/10 · Final project: 10/10

PROJECTS

Signal Watch – ML model performance monitoring (Master's thesis, 10/10) Python · Streamlit · pytest

- Built a monitoring system that detects performance decay in financial models using sequential change detection (CUSUM, Page-Hinkley), with the false-alarm rate calibrated per model.
- Validated on a synthetic credit-model test bench (1.28M observations, 200 replicas per scenario) and on 60+ years of real market data.
- Confirmed a realistic decay 17 months earlier than a single-observation baseline (38 vs 55 months) at the same false-alarm rate, and measured how calibration drifts out of sample.
- Shipped a Streamlit dashboard with PDF model-risk reports, 130 automated tests, CI and full reproducibility.

QuantAsistente – RAG assistant for quantitative finance LangChain · LangGraph · ChromaDB · Gemini

- Developed a retrieval-augmented LLM assistant over ~160 pages of finance literature (~300 chunks), with source and page citations in every answer.
- Added conversational memory with LangGraph and deployed it as a Streamlit web app on Streamlit Community Cloud.

Customer repurchase probability model Python · CatBoost · scikit-learn

- Trained a gradient-boosting model to predict each customer's probability of buying again (Online Retail II), and segmented customers for retention actions.

NLP analysis of customer reviews Python · NLP

- Analysed easyJet Trustpilot reviews with NLP to identify the drivers of customer satisfaction.

LendingClub credit risk analytics PostgreSQL · SQL

- Designed a layered SQL pipeline (staging, fact/dimension, views) for default-risk analysis, with data-quality checks, an audit-log trigger and a risk-scoring function.

EXPERIENCE

AI Reviewer (LLM evaluation) Dec 2024 – Feb 2026

- Evaluated LLM-generated outputs for accuracy, reasoning and guideline adherence, flagging factual and logical errors.
- Delivered structured feedback used to improve model responses in performance-driven evaluation projects.

Freelance Backend Developer Oct 2024 – Feb 2025

- Designed the database schema and core backend of a marketplace platform: authentication, role management and product listings.

Junior Software Developer – Backend & Automation Jan 2024 – Jul 2024

- Automated business workflows with Python and n8n, reducing manual work across company processes.
- Built MVC web applications (PHP, CodeIgniter, MySQL), integrated and tested REST APIs (Postman), and containerised environments with Docker.

SKILLS

Programming: Python, SQL, Java, Kotlin, C#, PHP, JavaScript, TypeScript

Data Science & ML: scikit-learn, CatBoost, pandas, NumPy, statistical inference, regression, time series, feature engineering, model evaluation, model monitoring, deep learning and neural networks (coursework)

LLMs & GenAI: RAG, LangChain, LangGraph, ChromaDB, vector search, prompt design, LLM evaluation

MLOps & Tools: Git, GitHub, CI, pytest, Docker, Streamlit, REST APIs, n8n, PostgreSQL, MySQL

Certifications: Google Cloud Data Analytics · Azure AI MLOps Engineer Associate (AI-300, in progress)

Currently learning: PyTorch, TensorFlow, MLflow, Azure Machine Learning, AWS, Apache Spark, Power BI

Languages: Spanish (native), Arabic (native), English (fluent), German (basic, currently learning)